

Lecture 3

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1 Erasure Correction

1.1 Only Erasure Correction

Claim: An (n, M, d) can correct upto **$d-1$** erasures**Proof:** (By contradiction) Note that channel only introduces erasures & no errors

$$D(y) = \begin{cases} c & \text{if } c \text{ is the unique codeword that 'agrees' with } y \\ \text{error} & \text{otherwise} \end{cases}$$

Contradiction: If we have c and c' such that $c' \neq c$ and both agree to y

$$\Rightarrow d(c, c') \leq d-1 \Rightarrow \text{Contradiction}$$

1.2 Error and Erasure Correction and Detection

For an (n, M, d) code over a channel introducing both errors and erasures, we have

$$2t + l + \rho \leq d - 1$$

Here:

- ρ is the number of erasures ($0 \leq \rho \leq d-1$)
- If # errors $\leq t$, then recovery possible
- If # errors $\leq t+l$, then errors will be detected

This analysis was possible since the erasures can be excluded and we can do a similar analysis as we did before for **Error correction and Detection**

2 Groups

Definition: A **group** is a set G along with an operation $*$ which acts on pairs of elements on G , satisfies the following properties:

- **Closure:** $\forall a, b \in G, a * b \in G$
- **Associativity:** $\forall a, b, c \in G, (a * b) * c = a * (b * c)$
- **Identity:** $\exists e \in G$ s.t. $a * e = e * a = a \quad \forall a \in G$
- **Inverse:** $\forall a \in G, \exists a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$
- **Commutativity*:** $\forall a, b \in G, a * b = b * a$ (Only for Abelian Groups)

Definition: $\text{Order}(G) = \# \text{ elements in } G$

2.1 Subgroups

Definition: A **subgroup** H of G satisfies the following properties

- H is closed under $*$
- $\forall a \in H, a^{-1} \in H$

2.2 Subgroups Construction

Let G be a finite group. Given an element $h \in G$, consider the following set:

$$H = \{h^j \mid j \in \mathbf{N}\}$$

Note that H is a finite set because every element of H belongs to the finite group G . This means that powers of h repeat after a point.

Claim: First element of H that repeats is h

Proof: We'll prove this by contradiction. If possible, let's say the first element to be repeated is h^i , where $i \in \mathbf{N} - 1$. Hence, $\exists j > i$ such that $h^i = h^j$. Pre-multiplying by h^{-1} on both sides gives $h^{i-1} = h^{j-1}$. Now, as i cannot be 1, hence, $i-1 \geq 1$. This means h^{i-1} repeats before h^i . Therefore, we obtain a contradiction, and $i=1$, which means h is the first element that repeats.

Let's say the set H is:

$$H = \{h, h^2, h^3, \dots, h^{j-1} = e\}$$

We say that $j-1$ is the order of element h , and H is called the subgroup spanned by the element h .

Note that this is a very specific form of a subgroup that is generated by powers of an element of G . A Subgroup of this form is called **Cyclic Subgroup**. There do exist subgroups that are not of this form.

2.3 Coset Decomposition

Let G be a finite group of order n . Let H be one of its Subgroups (H need not be a cyclic subgroup).

Let $G_1 = H$.

Take $g_2 \in G$ such that $g_2 \notin G_1$. Let $G_2 = g_2 \cdot H$ where $g_2 \cdot H$ is the set formed by pre-multiplying each element of H by g_2 .

Take $g_3 \in G$ such that $g_3 \notin G_1 \cup G_2$. Let $G_3 = g_3 \cdot H$.

Continue doing this till there's no element of G that's not present in any of the G_i 's. Let's say we form s such sets.

$$\begin{bmatrix} h_1 & h_2 & \cdots & h_m \\ g_2 * h_1 & g_2 * h_2 & \cdots & g_2 * h_m \\ \vdots & \vdots & \ddots & \vdots \\ g_s * h_1 & g_s * h_2 & \cdots & g_s * h_m \end{bmatrix}$$

Claim: Each element of G appears exactly once in the above decomposition.

Proof: The proof consists of 3 parts:

- No 2 elements in the same row are repeated:

We can prove this by contradiction. If possible, let's say $\exists i$ such that $g_i * h_j = g_i * h_k$, where $j \neq k$. Pre-multiplying by g_i^{-1} gives $h_j = h_k$, but this isn't possible as $j \neq k$. Therefore we obtain a contradiction to our assumption that such a g_i exists and hence No 2 elements in the same row are repeated.

- No 2 elements of different rows are repeated:

Again, we use contradiction. If possible, let's say $\exists a, b$ such that $a > b$ and $g_a * h_j = g_b * h_k$. Post multiplying by h_j^{-1} on both sides gives $g_a = (g_b * h_k) * h_j^{-1} = g_b * (h_k * h_j^{-1})$. Now, both $h_k, h_j^{-1} \in H$, which means their product also $\in H$. Now, the equation says $g_b * h_m$ is equal to g_a which means g_a lies in G_b . As $a > b$, this contradicts how we have formed cosets. Hence, No 2 elements of different rows are repeated.

- Every element of G is present in one of the G_i 's:

Note that we keep on forming cosets till there's no element of G that doesn't occur in the already existing cosets. And this would terminate in finite cosets because G is a finite set and every coset is formed by a distinct element of G .

Now, the above theorem implies that we have partitioned the group G into cosets of equal cardinality, equal to the order of subgroup H . If the order of G is n , the order of H is m , and the no of cosets is s , we clearly have $s \times m = n$. This means that the order of any subgroup H of a group G divides the order of G .

3 Fields

Definition: A field F is a set of elements with two operations: addition $(+)$ and multiplication $(\cdot)$, which satisfy:

- F is an Abelian group under $(+)$ with additive identity as 0
- F is closed under $(\cdot)$ and non-zero elements form an Abelian group under $(\cdot)$ with multiplicative identity as 1
- Distributive law holds $[a \cdot (b + c) = a \cdot b + a \cdot c]$

Example: $GF(2)$: $\{0, 1\}$ with mod-2 addition and mod-2 multiplication

$\oplus$	0	1
0	0	1
1	1	0

Addition

$\cdot$	0	1
0	0	0
1	0	1

Multiplication